

# Siemens LOGO! PLC — Ladder Logic Specification

 **Print-ready PDF:** <docs/pdf/plc-ladder-logic.pdf>

This document specifies all function blocks for the LOGO! Soft Comfort program. Program the LOGO! from this spec before connecting the rig to the RPi HMI.

## LOGO! Model

**LOGO! 12/24RCE** (6ED1052-1MD08-0BA1) - Supply: 12/24V DC - Inputs: 8 digital (24V DC), 4 analog - Outputs: 4 relay (250V AC, 10A each) - Expansion: up to 24 I/O via expansion modules - Ethernet: built-in for LOGO! Web Server + Modbus TCP (port 502) - Software: LOGO! Soft Comfort v8.x (included on CD)

## I/O Assignment

| Terminal | Type   | Signal name    | Description                        | Switch type |
|----------|--------|----------------|------------------------------------|-------------|
| I1       | Input  | ESTOP          | E-stop mushroom button             | NC          |
| I2       | Input  | DOOR_INTERLOCK | Chamber door reed switch           | NC          |
| I3       | Input  | OVERTEMP_HW    | 120°C bimetallic thermostat cutout | NO          |
| I4       | Input  | MANUAL_START   | Green momentary start button       | NO          |
| I5       | Input  | MANUAL_STOP    | Red momentary stop button          | NC          |
| I6–I8    | Input  | SPARE          | Reserved for future expansion      | —           |
| Q1       | Output | HEATER_RELAY   | SSR control — heater circuit       | Relay NO    |
| Q2       | Output | SERVO_POWER    | 24V→5V BEC power enable for servos | Relay NO    |
| Q3       | Output | DUT_POWER      | DUT power enable relay             | Relay NO    |
| Q4       | Output | ALARM_BEACON   | External 24V alarm beacon          | Relay NO    |

## Modbus TCP Register Map

Enable Modbus server in LOGO! Soft Comfort: **Tools** → **Ethernet** → **Modbus server** → **Enable**.

### Software enable inputs (written by RPi — M marker coils)

The RPi writes these marker flags as "software requests". The ladder ANDs each one with the hardware safety gate before driving the physical relay. This ensures the E-stop, door interlock, and overtemp cutout always take precedence over software.

| Modbus address | Access | LOGO! variable | Description                                            |
|----------------|--------|----------------|--------------------------------------------------------|
| Coil 2         | R/W    | M3             | SW_HEATER — RPi software heater request → gates Q1     |
| Coil 3         | R/W    | M4             | SW_SERVO — RPi software servo-power request → gates Q2 |
| Coil 4         | R/W    | M5             | SW_DUT — RPi software DUT-power request → gates Q3     |

### Physical output read-back (Q coils — read only from RPi)

| Modbus address | Access | LOGO! variable | Description                             |
|----------------|--------|----------------|-----------------------------------------|
| Coil 8192      | R      | Q1             | Heater relay actual state (read-back)   |
| Coil 8193      | R      | Q2             | Servo power relay actual state          |
| Coil 8194      | R      | Q3             | DUT relay actual state                  |
| Coil 8195      | R      | Q4             | Alarm beacon actual state (ladder-only) |

### Physical input monitoring (discrete inputs — read only)

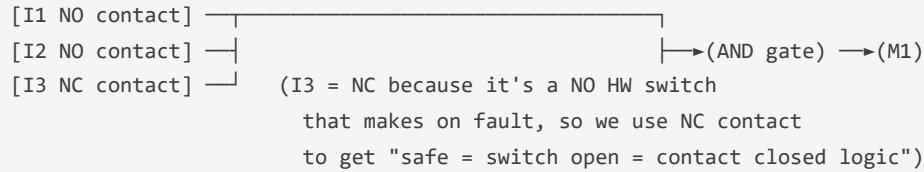
| Modbus address | Access | LOGO! variable | Description                             |
|----------------|--------|----------------|-----------------------------------------|
| Input 8192     | R      | I1             | E-stop physical state (1=safe)          |
| Input 8193     | R      | I2             | Door interlock (1=closed)               |
| Input 8194     | R      | I3             | HW overtemp (1=tripped)                 |
| Input 8195     | R      | I4             | Manual start button (1=pressed)         |
| Input 8196     | R      | I5             | Manual stop button (1=not-pressed=safe) |

# Function Block Program

## Network 1 — Safety gate (internal flag M1)

M1 is HIGH only when ALL physical safety conditions are met. M1 feeds into all output rungs.

FBD Network 1:



**Note:** I1 and I2 are NC switches → when safe, they provide 24V → LOGO! reads HIGH → use NO contact in ladder. I3 is NO switch → when tripped, provides 24V → use NC contact so M1 goes LOW when I3 is HIGH (tripped).

## Network 2 — Run flag SR latch (M2)

SET input: [I4 NO] (manual start button)  
RESET input: [I5 NC inverted] OR [M1 inverted] (stop button OR safety gate lost)  
Output: →(M2 = "run permitted")

## Network 3 — Q1 Heater relay

[M1 NO] —[M2 NO] —[M3 NO (SW\_HEATER)]→(Q1)

Heater energises only when: safety OK AND run latched AND RPi software requests it (M3).

## Network 4 — Q2 Servo power relay

[M1 NO] —[M4 NO (SW\_SERVO)]→(Q2)

Servo power requires only the safety gate and software request (M4). No run latch needed — servos may be positioned during idle or cool-down.

## Network 5 — Q3 DUT power relay

[M1 NO] —[M2 NO] —[M5 NO (SW\_DUT)]→(Q3)

DUT power requires all three conditions like the heater. M5 is the software request written by the RPi.

## Network 6 — Q4 Alarm beacon

[I1 NC inverted] —OR—[I2 NC inverted] —OR—[I3 NO]→(Q4)

Alarm fires immediately on any safety fault, independent of all software state. Cannot be silenced by writing M markers — only clearing the hardware fault extinguishes it. - NOT(I1) = alarm when E-stop tripped (I1 LOW → inverted → HIGH → alarm) - NOT(I2) = alarm when door open (I2 LOW → inverted → HIGH → alarm) - I3 = alarm when HW overtemp active (I3 HIGH → alarm)

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## LOGO! Soft Comfort Setup Procedure

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1. Install LOGO! Soft Comfort from the CD included with the PLC.
  2. Open Soft Comfort → **File** → **New** → select **LOGO! 8 / 12/24RC**.
  3. Set the PC's network adapter to the same subnet as the LOGO! (default: 192.168.0.x).
  4. Connect USB programming cable (or Ethernet if your LOGO! model supports it).
  5. **Tools** → **Ethernet settings** → **IP address**: set to `192.168.1.100` (or match `rig_config.json`).
  6. **Tools** → **Ethernet** → **Modbus server**: enable, port 502.
  7. Enter the FBD program as specified in Networks 1–6 above.
  8. **File** → **Transfer to LOGO!** — wait for transfer confirmation.
  9. Set LOGO! to **RUN mode** via the front-panel button or Soft Comfort.
  10. Verify: press Manual Start (I4) → M2 should latch. Check with Soft Comfort online monitor.
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## Verification Checklist

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Before connecting the heater or DUT:

- [ ] LOGO! powered up, showing RUN on display
  - [ ] Modbus TCP responding on port 502 (test with `logo_driver.py ping()`)
  - [ ] I1 reads HIGH (E-stop button released, NC circuit closed)
  - [ ] I2 reads HIGH (door closed)
  - [ ] I3 reads LOW (HW thermostat under 120°C, NO contact open)
  - [ ] Pressing Manual Start → M2 latches → Q1/Q3 coils can be enabled via software
  - [ ] Pressing E-stop → I1 goes LOW → M1 goes LOW → Q1, Q3 de-energise immediately
  - [ ] Q4 alarm fires when E-stop pressed
  - [ ] Modbus write to Q1 coil with safety gate open → Q1 relay clicks
  - [ ] Modbus write to Q1 coil with E-stop pressed → Q1 does NOT energise
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